Logistic Regression

The ***logistic regression model*** applies to ***binary*** response variables. For instance, logistic regression can model

- A voter's choice in a U.S. presidential election (Democrat or Republican), with explanatory variables political ideology, annual income, education level, and religious affiliation.
- Whether a person uses illegal drugs (yes or no), with explanatory variables education level, whether employed, religiosity, marital status, and annual income.

Logistic Regression

- Multicategory versions of logistic regression can handle ordinal response variables and nominal response variables.
- ***Loglinear models*** describe association structure among a set of categorical response variables.

15.1 Logistic Regression

- For a binary response variable y, denote its two categories by 1 and 0, commonly referred to as *success* and *failure*.
- Models for binary data ordinarily assume a *binomial distribution* for the response variable.

Linear Probability Model

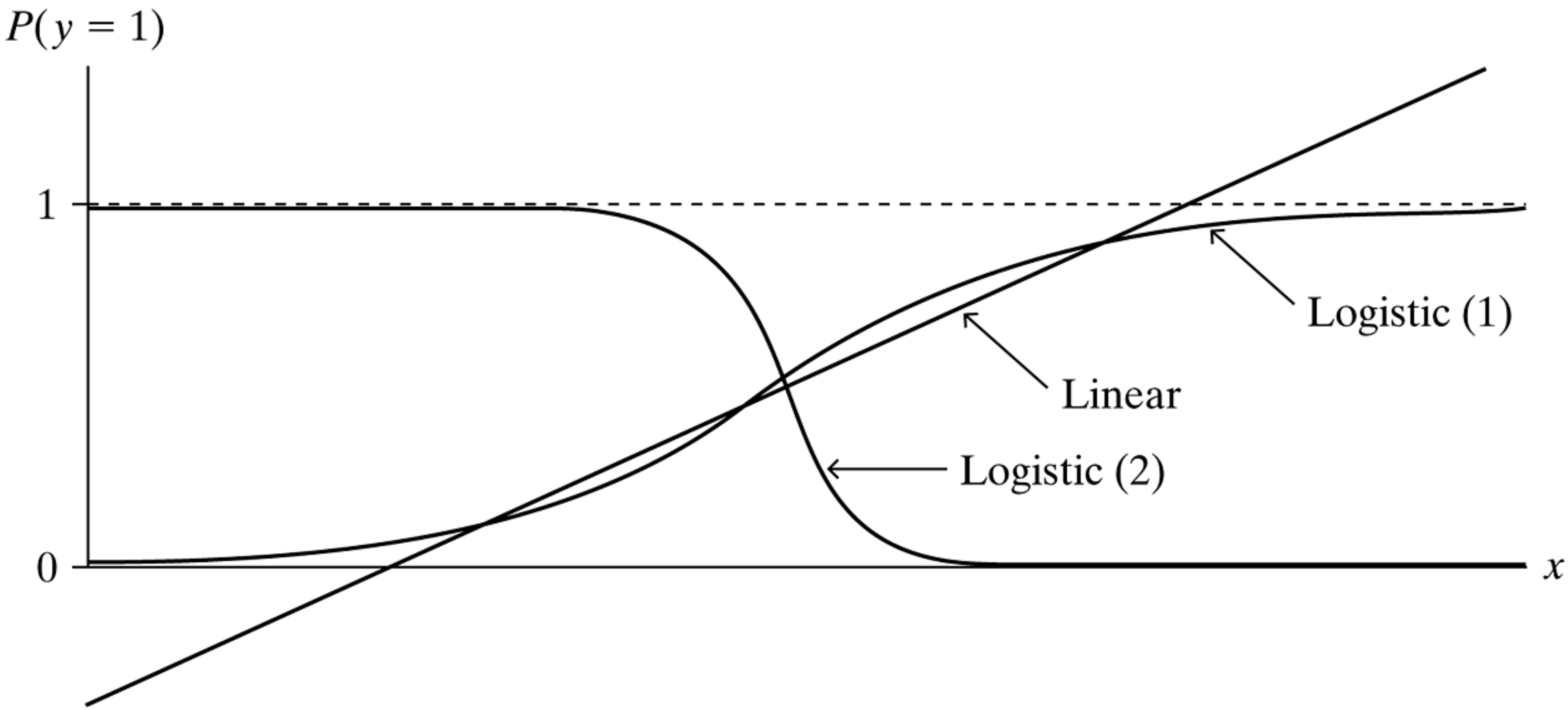
- For a single explanatory variable, the simple model

  $$P(y = 1) = \alpha + \beta x$$

  implies that the probability of success is a linear function of x.

  This is called the ***linear probability model***.

Linear Probability Model



Linear and Logistic Regression Models for a Binary (0, 1) Response, for Which $E(y) = P(y = 1)$

The Logistic Regression Model for Binary Responses

- The previous figure shows response curves which have an S-shape. The probability of a success falls between 0 and 1 for all possible *x* values.
- These relationships are described by the formula

$$\log\left[\frac{P(y=1)}{1-P(y=1)}\right] = \alpha + \beta x.$$

- The ratio $P(y = 1)/[1 - P(y = 1)]$ equals the ***odds***. For instance, when $P(y = 1) = 0.75$, the odds are $0.75/0.25 = 3.0$, meaning that a success is three times as likely as a failure.

The Logistic Regression Model for Binary Responses

- This formula uses the log of the odds, log $[P(y = 1)/(1 - P(y = 1))]$, called the ***logistic transformation***, or ***logit*** for short.
- The model, abbreviated as

$$\text{logit}[P(y = 1)] = \alpha + \beta x,$$

is called the ***logistic regression model***.

Example: Political Ideology and Belief in Evolution

The 2014 General Social Survey asked, “Human beings, as we know them today, developed from earlier species of animals. True or false?” Is the response to this question associated with one’s political ideology? Let y = opinion about evolution (1 = true, 0 = false). Let x = political ideology (1 = extremely conservative, 2 = conservative, 3 = slightly conservative, 4 = moderate, 5 = slightly liberal, 6 = liberal, 7 = extremely liberal). The next table shows 4 of the 1064 observations.

Example: Political Ideology and Belief in Evolution

TABLE 15.1: GSS Data on y = Opinion about Evolution (1 = True, 0 = False) and x = Political Ideology (from 1 = Extremely Conservative to 7 = Extremely Liberal)

Subject	x	y
1	4	1
2	3	0
3	2	0
4	6	1

Source: Complete data file `Evolution` for $n = 1064$ is at the text website.

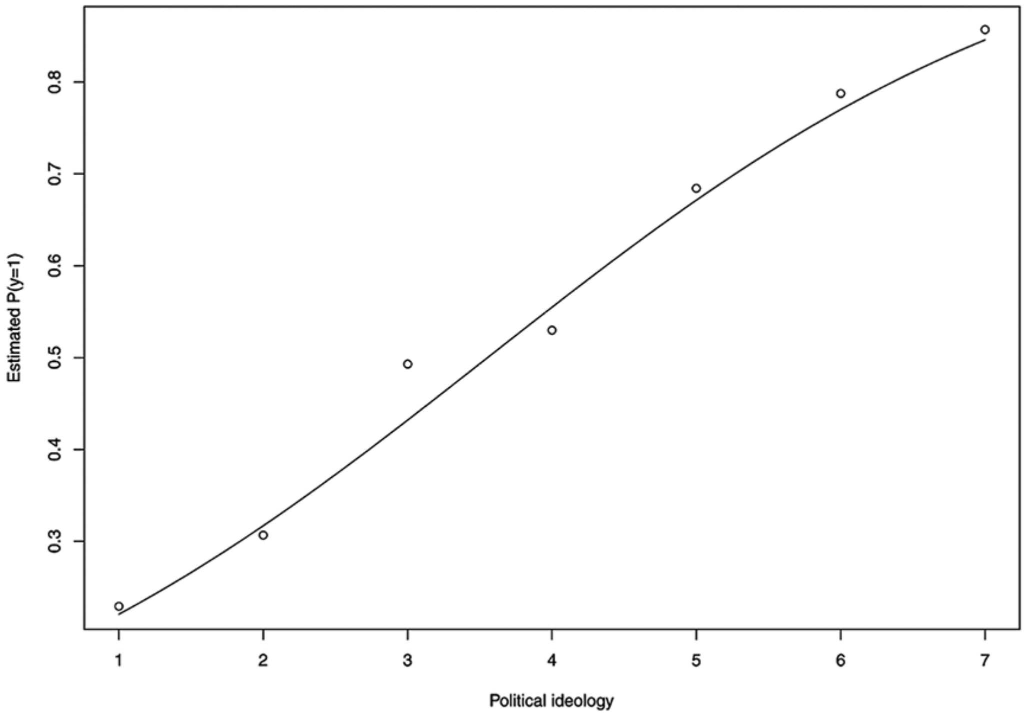
Example: Political Ideology and Belief in Evolution

Political ideology is an ordinal variable. As with any model, we can treat an ordinal explanatory variable in a quantitative manner if we expect a trend upward or a trend downward in y as x increases. We treat it as categorical with dummy variables for more general effects other than trends. For these data, if we expect that the probability of belief in evolution continually increases or continually decreases as a person is more liberal, we treat x as quantitative.

Example: Political Ideology and Belief in Evolution

The model is then more parsimonious and simpler to interpret than if we use dummy variables for categories of *x*. In fact, the sample proportion of responses in the *true* category, shown in the next figure, continually increases from 0.23 to 0.86 as political ideology moves from the most conservative to the most liberal. For a quantitative approach, it seems sensible to assign equally spaced scores for political ideology, such as the category numbers (1, 2, 3, 4, 5, 6, 7).

Example: Political Ideology and Belief in Evolution



Sample Proportions Believing in Evolution for Seven Political Ideology Categories, and Logistic Regression Prediction Curve for the Probability of Believing in Evolution

Example: Political Ideology and Belief in Evolution

The previous figure also shows the prediction curve for the fit of the logistic regression model. The next table shows some results that software provides for the model. The prediction equation is

$$\text{logit}[\hat{P}(y = 1)] = -1.757 + 0.494x.$$

Since $\hat{\beta} = 0.494 > 0$, the estimated probability of believing in evolution increases as political ideology moves in the more liberal direction (i.e., higher x scores). The estimated probability equals 0.50 at $x = -\hat{\alpha}/\hat{\beta} = 1.757/0.494 = 3.55$. The estimated probability of believing in evolution is below 0.50 for the three categories for which political ideology is conservative.

Example: Political Ideology and Belief in Evolution

TABLE 15.2: Logistic Regression Model Output (R Software) for the `Evolution` Data File on Belief in Evolution and Political Ideology

```
> summary(glm(y ~ polviews, family=binomial(link="logit"))

              Estimate   Std. Error   z value    Pr(>|z|)
(Intercept)   -1.75658      0.20500    -8.569    <2e-16
polviews       0.49422      0.05092     9.706    <2e-16
```

Logistic Regression Equation for Probabilities

- An alternative equation for logistic regression expresses the probability of success directly. It is

$$P(y = 1) = \frac{e^{\alpha+\beta x}}{1 + e^{\alpha+\beta x}}.$$

- Here, *e* raised to a power represents the antilog of that number, using natural logs.
- We use this formula to estimate values of $P(y = 1)$ at particular predictor values.

Logistic Regression Equation for Probabilities

- From the previous example, a person with political ideology x has estimated probability of believing in evolution

$$\hat{P}(y = 1) = \frac{e^{-1.757+0.494x}}{1 + e^{-1.757+0.494x}}.$$

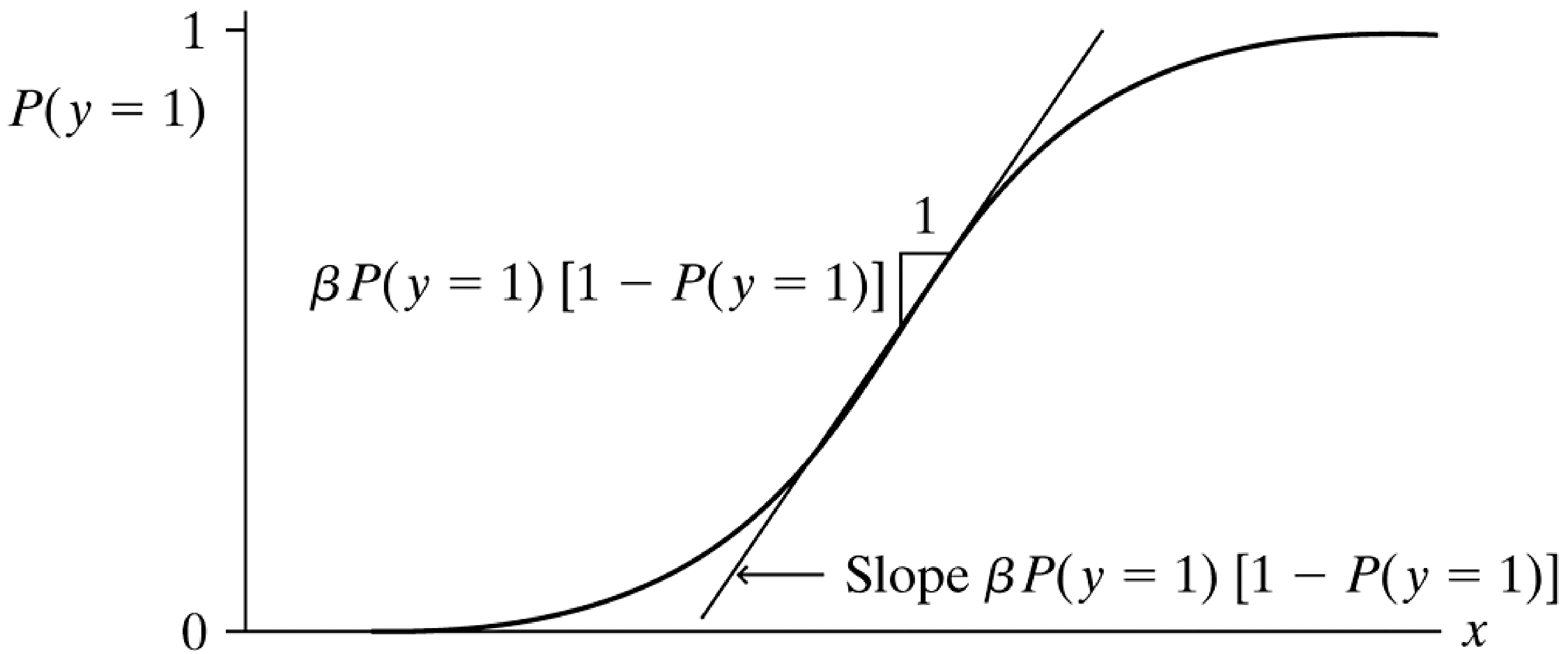
- For subjects with ideology $x = 1$, the most conservative category, the estimated probability equals

$$\hat{P}(y = 1) = \frac{e^{-1.757+0.494(1)}}{1 + e^{-1.757+0.494(1)}} = \frac{e^{-1.262}}{1 + e^{-1.262}} = 0.221.$$

Interpreting the Logistic Regression Model

- The simplest way to use β to interpret the steepness of the curve uses a straight-line approximation to the logistic regression curve.
- A straight line drawn tangent to the curve at a particular x-value has slope $\beta P(y = 1)[1 - P(y = 1)]$, where $P(y = 1)$ is the probability at that x.

Interpreting the Logistic Regression Model



A Line Drawn Tangent to a Logistic Regression Curve Has Slope $\beta P(y = 1)[1 - P(y = 1)]$, Which Is about $\beta/4$ when $P(y = 1)$ Is near 1/2

Interpreting the Logistic Regression Model

- For the political ideology and belief in evolution data, $\hat{\beta} = 0.494$. At $x = 4$ (*moderate* ideology), the estimated probability of believing in evolution is $\hat{P}(y = 1) = 0.555$. A line drawn tangent to the curve at that point has slope approximately equal to $\hat{\beta}/4 = 0.494/4 = 0.12$.
- So, a one-category increase in political ideology (i.e., from "moderate" to "slightly liberal") has approximately a 0.12 increase in the estimated probability of belief in evolution.

Interpretation Using the Odds and Odds Ratio

- Another interpretation of the logistic regression parameter β uses the *odds ratio*.
- Applying antilogs to both sides of the logistic regression equation $\log[P(y = 1)/(1 - P(y = 1))] = \alpha + \beta x$ yields the model expressed in terms of the *odds*,

$$\frac{P(y = 1)}{1 - P(y = 1)} = e^{\alpha+\beta x} = e^{\alpha}(e^{\beta})^{x}.$$

Interpretation Using the Odds and Odds Ratio

- In the previous example, the antilog of $\widehat{\beta}$ is $e^{\hat{\beta}} = e^{0.494} = 1.64$. When political ideology increases by one category in the liberal direction, the estimated odds of belief in evolution multiply by 1.64; that is, they increase by 64%. When $x = 5$, the estimated odds of belief in evolution are 1.64 times what they are when $x = 4$.
- When $x = 4$,

$$\text{Estimated odds} = \frac{\widehat{P}(y=1)}{1-\widehat{P}(y=1)} = e^{-1.757+0.494(4)}$$

$$= 1.246$$

Interpretation Using the Odds and Odds Ratio

- Whereas when $x = 5$,

$$\text{Estimated odds} = \frac{\hat{P}(y=1)}{1-\hat{P}(y=1)} = e^{-1.757+0.494(5)}$$

$$= 2.043,$$

which is 1.64 times the value of 1.246 at $x = 4$.

- In other words, $e^{\hat{\beta}} = e^{0.494} = 1.64 = 2.043/1.246$ is an estimated *odds ratio*, equaling the estimated odds at $x = 5$ divided by the estimated odds at $x = 4$.

Logistic Regression Can Use Grouped or Ungrouped Data Files

- The first example showed the data file in the usual form of one row of data for each subject.
- When the explanatory variables are categorical and the data can take the form of a contingency table, an alternative data file has a row of data for each cell count.
- The data are then said to be *grouped* instead of *ungrouped*.

15.2 Multiple Logistic Regression

- Logistic regression can handle multiple predictors. The multiple logistic regression model has the form

$$\text{logit}[P(y=1)] = \alpha + \beta_1 x_1 + \cdots + \beta_p x_p.$$

- The formula for the probability itself is

$$P(y=1) = \frac{e^{\alpha+\beta_1 x_1+\cdots+\beta_p x_p}}{1+e^{\alpha+\beta_1 x_1+\cdots+\beta_p x_p}}.$$

Example: Death Penalty and Racial Predictors

The table on the next slide is a three-dimensional contingency table from a study of the effects of racial characteristics on whether individuals convicted of homicide receive the death penalty. The variables in the table are *death penalty verdict*, the response variable, having categories (yes, no), and the explanatory variables *race of defendant* and *race of victims*, each having categories (white, black). The 674 subjects were defendants in indictments involving cases with multiple murders in Florida.

Example: Death Penalty and Racial Predictors

TABLE 15.3: Death Penalty Verdict by Defendant's Race and Victims' Race, for Cases with Multiple Murders in Florida

Defendant's Race	Victims' Race	Death Penalty Yes	Death Penalty No	Percentage Yes
White	White	53	414	11.3
	Black	0	16	0.0
Black	White	11	37	22.9
	Black	4	139	2.8

Example: Death Penalty and Racial Predictors

For each of the four combinations of defendant's race and victims' race, the previous table also lists the percentage of defendants who received the death penalty. For white defendants, the death penalty was imposed 11.3% of the time when the victims were white and 0.0% of the time when the victims were black, a difference of 11.3% − 0.0% = 11.3%. For black defendants, the death penalty was imposed 22.9% − 2.8% = 20.1% more often when the victims were white than when the victims were black. Thus, controlling for defendant's race by keeping it fixed, the percentage of *yes* death penalty verdicts was considerably higher when the victims were white than when they were black.

Example: Death Penalty and Racial Predictors

Now, consider the association between defendant's race and the death penalty verdict, controlling for victims' race. When the victims were white, the death penalty was imposed 22.9% − 11.2% = 11.7% more often when the defendant was black than when the defendant was white. When the victims were black, the death penalty was imposed 2.8% more often when the defendant was black than when the defendant was white. In summary, controlling for victims' race, black defendants were somewhat more likely than white defendants to receive the death penalty.

Example: Death Penalty and Racial Predictors

For y = death penalty verdict, let $y = 1$ denote the *yes* verdict. Since defendant's race and victims' race each have two categories, a single dummy variable can represent each. Let d be a dummy variable for defendant's race and v a dummy variable for victims' race, where

$d = 1$, defendant = white, $d = 0$, defendant = black,

$v = 1$, victims = white, $v = 0$, victims = black.

The logistic model with main effects for these explanatory variables is

$$\text{logit}[P(y = 1)] = \alpha + \beta_1 d + \beta_2 v.$$

Example: Death Penalty and Racial Predictors

Here, e^{β_1} is the odds ratio between the response variable and defendant's race, controlling for victims' race, and e^{β_2} is the odds ratio between the response and victims' race, controlling for defendant's race.

The next table shows software output for the model fit. The prediction equation is

$$\text{logit}[\hat{P}(y = 1)] = -3.596 - 0.868d + 2.404v.$$

Example: Death Penalty and Racial Predictors

Since *d* = 1 for white defendants, the *negative* coefficient of *d* means that the estimated odds of receiving the death penalty are *lower* for white defendants than for black defendants. Since *v* = 1 for white victims, the *positive* coefficient of *v* means that the estimated odds of receiving the death penalty are *higher* when the victims were white than when they were black.

TABLE 15.4: Parameter Estimates for Logistic Model for Death Penalty Data (Stata Output). Race dummy variables (*def* and *vic*) are coded as 1 for white and 0 for black.

y	Coef.	Std. Err.	z	P>|z|	[95% Conf.	Interval]
def	-.8677969	.3670742	-2.36	0.018	-1.587	-.1483
vic	2.404444	.600616	4.00	0.000	1.227	3.582
_cons	-3.596104	.5069138	-7.09	0.000	-4.590	-2.603

Example: Death Penalty and Racial Predictors

The antilog of $\widehat{\beta}_1$, namely, $e^{\hat{\beta}_1} = e^{-0.868} = 0.42$, is the estimated odds ratio between defendant's race and the death penalty, controlling for victims' race. The estimated odds of the death penalty for a white defendant equal 0.42 times the estimated odds for a black defendant. We list *white* before *black* in this interpretation, because the dummy variable was set up with $d = 1$ for *white* defendants. If we instead let $d = 1$ for black defendants rather than white, then we get $\widehat{\beta}_1 = 0.868$ instead of -0.868. Then, $e^{0.868} = 2.38$, which is $1/0.42$; that is, the estimated odds of the death penalty for a black defendant equal 2.38 times the estimated odds for a white defendant, controlling for victims' race.

Example: Death Penalty and Racial Predictors

For victims' race, $e^{2.404} = 11.1$. Since $v = 1$ for *white* victims, the estimated odds of the death penalty when the victims were *white* equal 11.1 times the estimated odds when the victims were black, controlling for defendant's race. This is a very strong effect.

Example: Death Penalty and Racial Predictors

This model assumes that both explanatory variables are associated with the response variable, but with a lack of interaction: The effect of a defendant’s race on the death penalty verdict is the same for each victim’s race and the effect of victims’ race is the same for each defendant’s race. This means that the estimated odds ratio between each explanatory variable and the response variable takes the same value at each category of the other predictor. For instance, the estimated odds ratio of 11.1 between victims’ race and the death penalty is the same when the defendants were white as when the defendants were black.

Multiplicative Effects on Odds

- The parameter estimates for the logistic regression model are linear effects, but on the scale of the *log* of the odds.
- It is easier to understand effects on the *odds* scale than the *log odds* scale.
- The antilogs of the parameter estimates are multiplicative effects on the odds.

Multiplicative Effects on Odds

- To illustrate, for the data on the death penalty, the prediction equation

$$\text{Log}\left[\frac{\hat{P}(y=1)}{1-\hat{P}(y=1)}\right] = -3.596 - 0.868d + 2.404v$$

refers to the log odds (i.e., logit).

- The corresponding prediction equation for the estimated odds is

$$\text{Odds} = e^{-3.596-0.868d+2.404v} = e^{-3.596}e^{-0.868d}e^{2.404v}.$$

Multiplicative Effects on Odds

- The logit model expression for the log odds is *additive*, but taking antilogs yields a *multiplicative* expression for the odds.
- In other words, the antilogs of the parameters are *multiplied* to obtain odds.

Effects on Probabilities are Simpler to Interpret

- Many researchers find it easier to get a feel for the effects by viewing summaries that use the *probability* scale.
- Such summaries can report estimated probabilities at particular values of a variable of interest.

Effects on Probabilities are Simpler to Interpret

- The formula for the estimated probability of receiving the death penalty is

$$\hat{P}(y = 1) = \frac{e^{-3.596-0.868d+2.404v}}{1 + e^{-3.596-0.868d+2.404v}}.$$

- For instance, when the defendant is white and the victims were white, $d = v = 1$, so

$$\hat{P}(y = 1) = \frac{e^{-3.596-0.868(1)+2.404(1)}}{1 + e^{-3.596-0.868(1)+2.404(1)}} = \frac{e^{-2.059}}{1 + e^{-2.059}}$$
$$= 0.113.$$

Effects on Probabilities are Simpler to Interpret

- When the defendant is black and the victims were white, $d = 0$ and $v = 1$, so

$$\hat{P}(y=1) = \frac{e^{-3.596-0.868(0)+2.404(1)}}{1+e^{-3.596-0.868(0)+2.404(1)}} = 0.233.$$

- The estimated probability of receiving the death penalty is about twice as high for black defendants. The sample effect is quite strong.
- The probability modeled relates to the odds by

$$\hat{P}(y=1) = \frac{\text{Odds}}{1+\text{Odds}}$$

Effects on Probabilities are Simpler to Interpret

- With a quantitative explanatory variable x, you can report the change in $\widehat{P}(y = 1)$ at the means of the other explanatory variables when x increases by a certain amount, such as (1) by a fixed value (e.g., 1), (2) by a standard deviation, (3) over its range from its lowest to greatest value, or (4) over its interquartile range from the lower quartile to the upper quartile.
- Approach (4) is, unlike (1), not affected by the choice of scale and, unlike (2) and (3), not affected by outliers.

Standardized Effects

- You can compare regression parameter estimates after the model has been fitted using standardized explanatory variables.
- An estimate then refers to the effect on the logit of a standard deviation increase in the value of the explanatory variable. If s_{x_j} is the standard deviation of explanatory variable x_j , the standardized estimate is

$$\hat{\beta}^*_j = \hat{\beta}_j s_{x_j}.$$

15.3 Inference for Logistic Regression Models

- Statistical inference assumes randomization for gathering the data. It also assumes a *binomial* distribution for the response variable.
- The model identifies y as having a binomial distribution and uses the logit link function for $P(y = 1)$, which is the mean of y.

Inference for Logistic Regression Models

- As in ordinary regression modeling, the logistic regression model

$$\text{logit}[P(y = 1)] = \alpha + \beta_1 x_1 + \cdots + \beta_p x_p$$

has three types of hypotheses.

- The global null hypothesis $H_0: \beta_1 = \cdots = \beta_p = 0$ states that none of the explanatory variables has an effect on $P(y = 1)$.
- An individual null hypothesis such as $H_0: \beta_1 = 0$ states that x_1 has no effect on y, controlling for the other explanatory variables.

Likelihood-Ratio Test Comparing Logistic Regression Models

- In ordinary regression, we use an *F* test to compare nested models.
- For logistic regression, the analogous test is the ***likelihood-ratio test***.
- This is a general purpose test in statistical inference that provides a way to compare two models, a full model and a simpler model.

Likelihood-Ratio Test Comparing Logistic Regression Models

- Let ℓ_0 denote the maximum of the likelihood function when H_0 is true, and let ℓ_1 denote the maximum without that assumption.
- The formula for the likelihood-ratio test statistic is

$$-2\log\left(\frac{\ell_0}{\ell_1}\right) = (-2\log\ell_0) - (-2\log\ell_1).$$

Likelihood-Ratio and Wald Tests About Effects

- The likelihood-ratio test can also be used for an individual parameter such as H_0: $\beta_i = 0$. The estimate $\hat{\beta}$ of β has an approximate normal sampling distribution.
- So, another possible test for H_0: $\beta_i = 0$ uses as test statistic $z = \hat{\beta}_i / se$.
- Some software instead reports the square of this statistic, called a ***Wald statistic***.

Example: Inference for Death Penalty and Racial Predictors

For the death penalty data, the second example used the model

$$\text{logit}[P(y = 1)] = \alpha + \beta_1 d + \beta_2 v,$$

with dummy variables d and v for defendant's race and victims' race, respectively. Software (Stata) shows the results in the next table.

Example: Inference for Death Penalty and Racial Predictors

TABLE 15.6: Logistic Regression Inference (Stata Output) for Death Penalty Data of Table 15.3

```
                                             LR chi2(2)  =  21.89
                                             Prob > chi2 = 0.0000
    y |     Coef.   Std. Err.     z    P>|z|  [95% Conf. Interval]
  def | -.8677969  .3670742   -2.36   0.018 -1.587249   -.1483447
  vic |  2.404444   .600616    4.00   0.000  1.227258    3.581629
_cons | -3.596104  .5069138   -7.09   0.000 -4.589637   -2.602571
```

Example: Inference for Death Penalty and Racial Predictors

The likelihood-ratio (LR) of 21.89 shown at the top of the output is the test of H_0: $\beta_1 = \beta_2 = 0$, that neither defendant's race nor victims' race has an effect. With $p = 2$ parameters, it is a chi-squared statistic with $df = 2$, and has P-value = 0.0000.

If $\beta_1 = 0$, the death penalty verdict is independent of defendant's race, controlling for victims' race. The defendant's race effect $\hat{\beta}_1 = -0.868$ has a standard error of 0.367. The z test statistic for H_0: $\beta_1 = 0$ is $z = \hat{\beta}_1/se = -0.868/0.367 = -2.36$. For the two-sided alternative, the P-value is 0.018. Similarly, the test of H_0: $\beta_2 = 0$ has $P = 0.000$. The tests provide strong evidence of individual effects.

Example: Inference for Death Penalty and Racial Predictors

The parameter estimates are also the basis of confidence intervals for odds ratios. Since the estimates refer to *log* odds ratios, after constructing the interval for a β_j we take antilogs of the endpoints to form the interval for the odds ratio. For instance, since the estimated log odds ratio of 2.404 between victims' race and the death penalty verdict has a standard error of 0.601, a 95% confidence interval for the true log odds ratio is

$$2.404 \pm 1.96(0.601), \text{ or } (1.23, 3.58).$$

Example: Inference for Death Penalty and Racial Predictors

The confidence interval for the odds ratio is $(e^{1.23}, e^{3.58})$ = (3.4, 35.9). For a given defendant's race, when the victims were white, the estimated odds of the death penalty are between 3.4 and 35.9 times the estimated odds when the victims were black.

Most software can also provide confidence intervals for probabilities. Ninety-five percent confidence intervals for the probability of the death penalty are (0.14, 0.37) for black defendants with white victims, (0.01, 0.07) for black defendants with black victims, (0.09, 0.15) for white defendants with white victims, and (0.003, 0.04) for white defendants with black victims.

15.4 Logistic Regression Models for Ordinal Variables

- Many applications have a categorical response variable with more than two categories.
- An extension of logistic regression can handle ordinal response variables.

Cumulative Probabilities and Their Logits

- For an ordinal response variable y, let $P(y \leq j)$ denote the probability that the response falls in category j or below.
- This is called a ***cumulative probability***.
- With four categories, for example, the cumulative probabilities are

$$P(y = 1),\ P(y \leq 2) = P(y = 1) + P(y = 2),$$

$$P(y \leq 3) = P(y = 1) + P(y = 2) + P(y = 3),$$

and the final cumulative probability uses the entire scale, so $P(y \leq 4) = 1$.

Cumulative Probabilities and Their Logits

- A c-category response has c cumulative probabilities. The order of forming the cumulative probabilities reflects the ordering of the response scale. The probabilities satisfy

$$P(y \leq 1) \leq P(y \leq 2) \leq \cdots \leq P(y \leq c) = 1.$$

- The odds of response in category j or below is the ratio

$$\frac{P(y \leq j)}{P(y > j)}.$$

Cumulative Probabilities and Their Logits

- A popular logistic model for an ordinal response variable uses logits of the cumulative probabilities. With c = 4, the logits are

$$\text{logit}[P(y \le 1)] = \log\left[\frac{P(y=1)}{P(y>1)}\right] = \log\left[\frac{P(y=1)}{P(y=2)+P(y=3)+P(y=4)}\right],$$

$$\text{logit}[P(y \le 2)] = \log\left[\frac{P(y\le 2)}{P(y>2)}\right] = \log\left[\frac{P(y=1)+P(y=2)}{P(y=3)+P(y=4)}\right],$$

$$\text{logit}[P(y \le 3)] = \log\left[\frac{P(y\le 3)}{P(y>3)}\right] = \log\left[\frac{P(y=1)+P(y=2)+P(y=3)}{P(y=4)}\right].$$

- The final cumulative probability equals 1.0.
- These logits of cumulative probabilities are called ***cumulative logits***.

Cumulative Logit Models for an Ordinal Response

- A model can simultaneously describe the effect of explanatory variables on all the cumulative probabilities for y.
- For each cumulative probability, the model looks like an ordinary logistic regression, where the two outcomes are low = "category j or below" and high ="above category j."
- With a single explanatory variable, this model is

$$\text{logit}[P(y \leq j)] = \alpha_j - \beta x, \quad j = 1, 2, \ldots, c - 1.$$

Cumulative Logit Models for an Ordinal Response

- Software (such as SAS and the *VGAM* library in R) that specifies the model as

$$\text{logit}[P(y \leq j)] = \alpha_j + \beta x$$

will report the opposite sign for $\hat{\beta}$. You should be careful to check how your software defines the model so that you interpret effects properly.

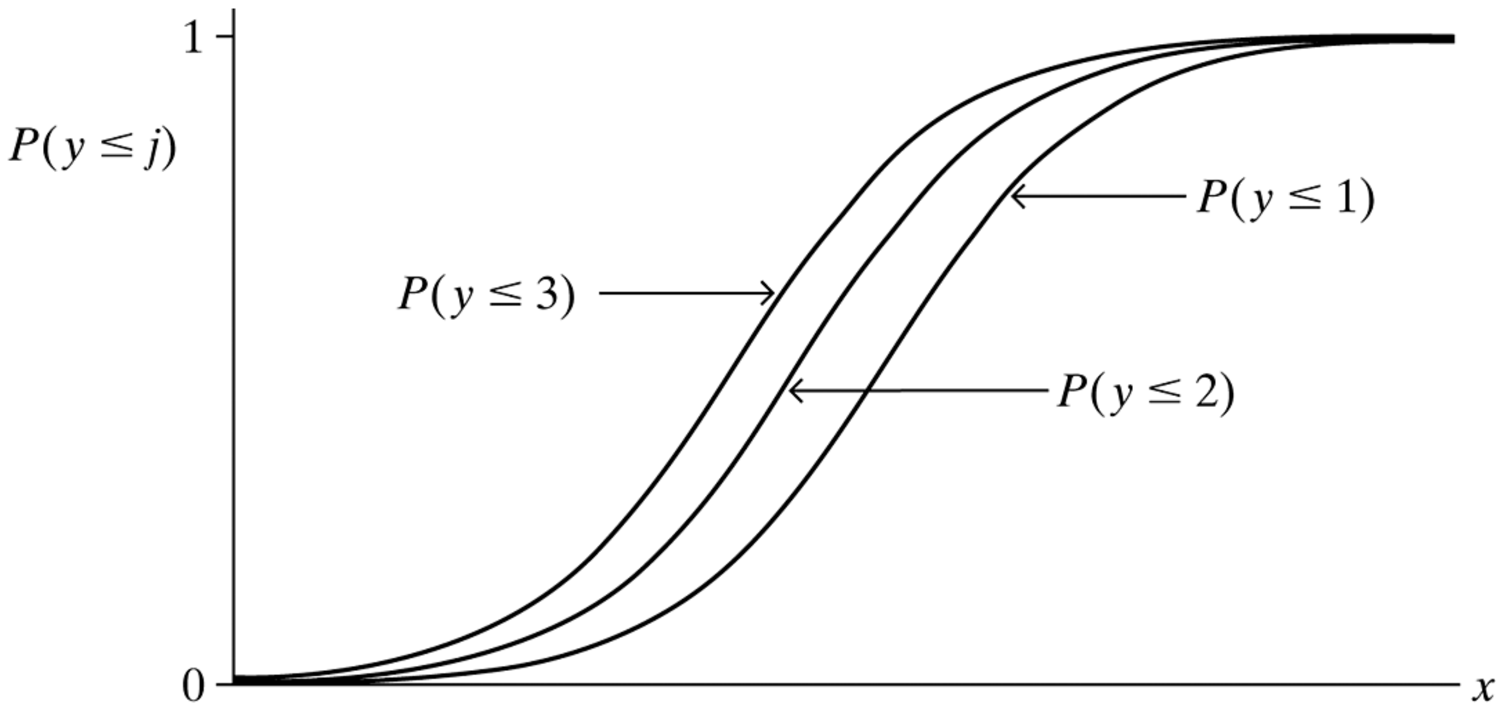
Cumulative Logit Models for an Ordinal Response

- The parameter β describes the effect of x on y. When $\beta = 0$, each cumulative probability does not change as x changes, and the variables are independent. The effect of x increases as $|\beta|$ increases.
- β is the same for each cumulative probability.
- This cumulative logit model with this common effect is often called the ***proportional odds*** model.

Cumulative Logit Models for an Ordinal Response

- The figure on the next slide depicts the model for four response categories with a quantitative explanatory variable.
- The model implies a separate S-shaped curve for each of the three cumulative probabilities.

Cumulative Logit Models for an Ordinal Response



Depiction of Curves for Cumulative Probabilities in a Cumulative Logit Model for a Response Variable with Four Categories

Cumulative Logit Models for an Ordinal Response

- The size of $|\beta|$ determines how quickly the curves climb or drop.
- The common value for β means that the three response curves have the same shape.
- Model fitting treats the observations as independent from a *multinomial distribution*, the generalization of the binomial distribution from two outcome categories to multiple categories.

Cumulative Logit Models for an Ordinal Response

- Cumulative logit models can handle multiple explanatory variables, which can be quantitative and/or categorical. The model has the form

$$\text{logit}[P(y \leq j)] = \alpha_j - \beta_1 x_1 - \beta_2 x_2 - \cdots - \beta_p x_p, \quad j = 1,2, \ldots, c - 1.$$

- When an explanatory variable is categorical, dummy variables can represent the categories.
- The model has the proportional odds assumption that the effect of an explanatory variable is the same on each cumulative probability.

Example: Comparing Political Ideology of Democrats and Republicans

In the United States, do Republicans tend to be more conservative than Democrats? The next table, for subjects in the 2014 General Social Survey, relates political ideology (EC = extremely conservative, C = conservative, SC = slightly conservative, M = moderate, SL = slightly liberal, L = liberal, EL = extremely liberal) to political party affiliation. We treat political ideology as the response variable. In the first example of this chapter, we treated it as a quantitative explanatory variable by assigning scores to categories. Now, we treat it as an ordinal response variable in a cumulative logit model.

Example: Comparing Political Ideology of Democrats and Republicans

TABLE 15.7: Political Ideology by Political Party Affiliation

Political Party	Political Ideology						
	EC	C	SC	M	SL	L	EL
Democratic	16	40	73	330	126	167	60
Republican	59	206	112	124	18	12	2

Example: Comparing Political Ideology of Democrats and Republicans

Let x be a dummy variable for political party affiliation, with $x = 1$ for Democrats and $x = 0$ for Republicans. The table on the next slide shows results of fitting the cumulative logit model. The response variable (political ideology) has seven categories, so the table reports six intercept parameter estimates, referred to as *cuts* because of how their ordered values are cutpoints on the real line. These estimates are not as relevant as the estimated effect of the explanatory variable (party affiliation), which is $\hat{\beta} = 2.527$.

Example: Comparing Political Ideology of Democrats and Republicans

TABLE 15.8: Output (Stata) for Cumulative Logit Model Fitted to Table 15.7

```
. ologit response party

Ordered logistic regression                        LR chi2(1)  = 506.60
                                                   Prob > chi2 = 0.0000
------------------------------------------------------------------------
 ideology |     Coef.    Std. Err.      z    P>|z|   [95% Conf. Int.]
   party  |   2.527265   .1224756   20.63   0.000    2.2872    2.7673
   /cut1  |  -1.969594   .1249231                   -2.2144   -1.7247
   /cut2  |  -.0430447    .083657                   -.20701    .12092
   /cut3  |   .8684324   .0896656                    .69269    1.0442
   /cut4  |   2.781418   .1172596                    2.5516    3.0112
   /cut5  |   3.481216   .1264142                    3.2334    3.7290
   /cut6  |   5.077736   .1690398                    4.7464    5.4090
```

Example: Comparing Political Ideology of Democrats and Republicans

Since the dummy variable *x* is 1 for Democrats and since high values of *y* represent greater liberalism, the *positive* $\hat{\beta}$ -value means that Democrats tend to be *more* liberal than Republicans. Democrats are more likely than Republicans to fall toward the liberal end of the political ideology scale.

Example: Comparing Political Ideology of Democrats and Republicans

We can also interpret $\hat{\beta} = 2.527$ by exponentiating $\hat{\beta}$ to form an estimated odds ratio using cumulative probabilities. For any fixed *j*, the estimated odds that a Democrat's response is in the liberal direction rather than the conservative direction (i.e., $y > j$ rather than $y \leq j$) are $e^{\hat{\beta}} = e^{2.527} = 12.5$ times the estimated odds for Republicans. Specifically, this odds ratio applies to each of the six cumulative probabilities for the first table in the example.

Example: Comparing Political Ideology of Democrats and Republicans

To illustrate, using the third of the six cumulative probabilities, the estimated odds that a Democrat is in a liberal category, rather than a moderate or a conservative category, are 12.5 times the corresponding estimated odds for a Republican. The value 12.5 is far from the no-effect value of 1.0. The sample has a strong association, Democrats tending to be more liberal than Republicans.

Inference for Effects on an Ordinal Response

- When $\beta = 0$ in the cumulative logit model, the variables are independent. We test independence by testing H_0: $\beta = 0$.
- These tests of independence take into account the ordering of the response categories.
- With multiple explanatory variables, to check the fit of the model we can analyze whether extra terms such as interactions provide a significant improvement in the model fit.

Inference for Effects on an Ordinal Response

- Don't put too much emphasis on statistical tests, whether of effects or of goodness of fit. Results are sensitive to sample size, more significant results tending to occur with larger sample sizes.
- A confidence interval for the odds ratio describes the association for the cumulative logit model.

Invariance to Choice of Response Categories

- When the cumulative logit model fits well, it also fits well with similar effects for any collapsing of the response categories.
- The cumulative logit model implies trends upward or downward among distributions of y at different values of explanatory variables.

15.5 Logistic Models for Nominal Responses

- For nominal response variables (i.e., *unordered* categories), an extension of the binary logistic regression model provides an ordinary logistic model for each pair of response categories.
- The models simultaneously use all pairs of categories by specifying the odds of outcome in one category instead of another.
- The order of listing the categories is irrelevant, because the response scale is nominal.

Baseline-Category Logits

- Logit models for nominal response variables pair each category with a baseline category.
- Most software uses the last category as the baseline.
- With three response categories, for example, the *baseline-category logits* are

$$\log\left[\frac{P(y=1)}{P(y=3)}\right] \text{ and } \log\left[\frac{P(y=2)}{P(y=3)}\right].$$

Baseline-Category Logits

- For c outcome categories, the baseline-category logit model is

$$\log\left[\frac{P(y=j)}{P(y=c)}\right] = \alpha_j + \beta_j x, \quad j = 1, \ldots, c-1.$$

- Given that the response falls in category j or the last category, this models the log odds that the response is j.
- It looks like an ordinary logistic regression model, where the two outcomes are category j and category c.

Baseline-Category Logits

- Each of the $c - 1$ logit equations has its own parameters.
- With multiple explanatory variables, the model

$$\log\left[\frac{P(y=j)}{P(y=c)}\right] = \alpha_j + \beta_{j1}x_1 + \beta_{j2}x_2 + \cdots + \beta_{jp}x_p,$$

$$j = 1, \ldots, c-1,$$

can have a large number of parameters.

- Software for multicategory logit models fits all the equations *simultaneously*, assuming independent observations from multinomial distributions.

Example: Belief in Afterlife by Sex and Race

The table on the next slide, from a General Social Survey, has y = belief in life after death, with categories (yes, undecided, no), and explanatory variables sex and race. Let s = 1 for females and 0 for males, and r = 1 for blacks and 0 for whites. With *no* as the baseline category for y, the model is

$$\log\left[\frac{P(y=1)}{P(y=3)}\right] = \alpha_1 + \beta_{1S}\, s + \beta_{1R}\, r,$$

$$\log\left[\frac{P(y=2)}{P(y=3)}\right] = \alpha_2 + \beta_{2S}\, s + \beta_{2R}\, r.$$

The S and R subscripts identify the sex and race parameters.

Example: Belief in Afterlife by Sex and Race

For example, β_{1S} compares females and males (controlling for race) on the log odds of responding *yes* rather than *no* to belief in life after death, whereas β_{2S} compares females and males on the log odds of responding *undecided* instead of *no*.

TABLE 15.9: Belief in Afterlife by Sex and Race

		Belief in Afterlife		
Race	Sex	Yes	Undecided	No
Black	Female	64	9	15
	Male	25	5	13
White	Female	371	49	74
	Male	250	45	71

Example: Belief in Afterlife by Sex and Race

The model assumes a lack of interaction between sex and race in their effects on belief in life after death. The next table shows the parameter estimates. For the first equation, for the log odds of a *yes* rather than a *no* response on belief in the afterlife, $\hat{\beta}_{1S} = 0.419$ and $\hat{\beta}_{1R} = -0.342$. Since the dummy variables are 1 for females and for blacks, given that a subject's response was *yes* or *no*, the estimated probability of a *yes* response was higher for females than for males (given race) and lower for blacks than for whites (given sex).

Example: Belief in Afterlife by Sex and Race

The effect parameters represent log odds ratios with the baseline category. For instance, $\hat{\beta}_{1S} = 0.419$ is the conditional log odds ratio between sex and response categories 1 and 3 (*yes* and *no*), given race. For females, the estimated odds of response *yes* rather than *no* on life after death are $e^{0.419} = 1.52$ times those for males, controlling for race. For blacks, the estimated odds of response *yes* rather than *no* on life after death are $e^{-0.342} = 0.71$ times those for whites, controlling for sex. These odds ratios suggest that, conditional on the response being *yes* or *no*, females are more likely than males to respond *yes*, and whites are more likely than blacks to respond *yes*.

Example: Belief in Afterlife by Sex and Race

TABLE 15.10: Output (Stata) for Baseline-Category Logit Model Fitted to Table 15.9. The first equation uses belief categories (yes, no) and the second equation uses belief categories (undecided, no).

```
. mlogit afterlife sex race [fweight = count], base(3)

 afterlife |      Coef.   Std. Err.     z     P>|z|   [95% Conf. Int.]
1      sex |   .4185504    .171255    2.44    0.015    .0829    .7542
      race |  -.3417744   .2370375   -1.44    0.149   -.8064    .1228
     _cons |   1.224826   .1281071    9.56    0.000    .9737    1.476
           |
2      sex |   .1050638   .2465096    0.43    0.670   -.3781    .5882
      race |  -.2709753   .3541269   -0.77    0.444   -.9651    .4231
     _cons |  -.4870336   .1831463   -2.66    0.008   -.8460   -.1281
           |
3          |  (base outcome)
```

Arbitrary Pairs of Response Categories, and Fitted Response Distributions

- The choice of the baseline category is arbitrary.
- We can use the equations for a given choice to get equations for *any* pair of categories. For example, using properties of logarithms,

$$\log\left[\frac{P(y=1)}{P(y=2)}\right] = \log\left[\frac{P(y=1)}{P(y=3)}\right] - \log\left[\frac{P(y=2)}{P(y=3)}\right].$$

Arbitrary Pairs of Response Categories, and Fitted Response Distributions

- So, to get the prediction equation for the log odds of belief *yes* instead of *undecided*, we take

$$(1.225 + 0.419s - 0.342r) - (-0.487 + 0.105s - 0.271r) = 1.71 + 0.314s - 0.071r.$$

- For example, the odds of a response *yes* instead of *undecided* are higher for females (for whom $s = 1$) than males ($s = 0$).

Arbitrary Pairs of Response Categories, and Fitted Response Distributions

- Since the odds ratios for this model refer to *conditional probabilities*, given outcome in one of two categories, it is also useful to have software report the estimated unconditional probabilities based on the model fit.

Inference for Baseline-Category Logit Models

- Inference applies as in ordinary logistic regression, except now to test the effect of an explanatory variable we consider *all* its parameters for the various equations.
- Likelihood-ratio tests compare the fits of the models with and without the explanatory variable in the model.

Inference for Baseline-Category Logit Models

- For the data on belief in an afterlife, the test of the sex effect has H_0: $\beta_1^S = \beta_2^S = 0$.
- The likelihood-ratio test compares the model to the simpler model dropping sex as an explanatory variable.
- The test statistic equals 6.75. It has $df = 2$, since H_0 has two parameters. The P-value of 0.03 shows evidence of a sex effect.
- The effect of race is not significant, the likelihood-ratio statistic equaling 1.99 on $df = 2$.

Inference for Baseline-Category Logit Models

TABLE 15.11: Output (R) for Baseline-Category Logit Model Fitted to Table 15.9, Also Showing Fitted Probabilities

```
> race <- c(1,1,0,0); sex <- c(1,0,1,0)
> y1 <- c(64,25,371,250); y2 <- c(9,5,49,45); y3 <- c(15,13,74,71)
> library(VGAM)
> fit <- vglm(cbind(y1,y2,y3) ~ race + sex, family=multinomial)
> summary(fit)
               Estimate  Std. Error  z value  Pr(>|z|)
(Intercept):1    1.2248      0.1281    9.561   < 2e-16
(Intercept):2   -0.4870      0.1831   -2.659   0.00783
race:1          -0.3418      0.2370   -1.442   0.14934
race:2          -0.2710      0.3541   -0.765   0.44416
sex:1            0.4186      0.1713    2.444   0.01452
sex:2            0.1051      0.2465    0.426   0.66996
---
> fitted(fit)
         y1         y2        y3
1 0.7073517 0.10018119 0.1924671
2 0.6221640 0.12055943 0.2572766
3 0.7545608 0.09956287 0.1458763
4 0.6782703 0.12244794 0.1992817
```

15.6 Loglinear Models for Categorical Variables

- Logistic regression models are similar in structure to ordinary regression models, both types predicting a response variable using explanatory variables.
- ***Loglinear models*** are appropriate for contingency tables in which each classification is a response variable.
- Loglinear models are special cases of generalized linear models that assume that each cell count in a contingency table has a ***Poisson distribution***.

Example: Students' Use of Alcohol, Cigarette, and Marijuana

The table on the next slide is from a survey by the Wright State University School of Medicine and the United Health Services in Dayton, Ohio. The survey asked senior high school students in a nonurban area near Dayton, Ohio, whether they had ever used alcohol, cigarettes, or marijuana. The table is a $2 \times 2 \times 2$ contingency table that cross-classifies responses on these three items.

Example: Students' Use of Alcohol, Cigarette, and Marijuana

TABLE 15.12: Alcohol, Cigarette, and Marijuana Use for High School Seniors

Alcohol Use	Cigarette Use	Marijuana Use	
		Yes	No
Yes	Yes	911	538
	No	44	456
No	Yes	3	43
	No	2	279

Source: Thanks to Prof. Harry Khamis, Wright State University, for these data.

Example: Students' Use of Alcohol, Cigarette, and Marijuana

In this table, all three variables are response variables, rather than one being a response variable and the others explanatory. The models presented in this section describe their association structure. They analyze whether each pair of variables is associated and whether the association is the same at each category of the third variable.

A Hierarchy of Loglinear Models for Three Variables

- Loglinear models apply to contingency tables with any number of dimensions.
- Loglinear models describe ***conditional associations*** in partial tables that relate two of the variables while controlling for the third one.

A Hierarchy of Loglinear Models for Three Variables

- A pair of variables could be statistically independent at each category of the third variable.
- In other words, the population version of each partial table could satisfy independence. In that case, the variables are said to be ***conditionally independent***, and the odds ratios equal 1 in the partial tables.

A Hierarchy of Loglinear Models for Three Variables

We now introduce a hierarchy of five loglinear models, ordered in terms of the extent of association.

1. All three pairs of variables are conditionally independent. That is,

   x is independent of *y*, controlling for *z*;

   x is independent of *z*, controlling for *y*;

   y is independent of *z*, controlling for *x*.

A Hierarchy of Loglinear Models for Three Variables

2. Two of the pairs of variables are conditionally independent. For example,

   x is independent of *z*, controlling for *y*;

   y is independent of *z*, controlling for *x*;

   x and *y* are associated, controlling for *z*.

3. One of the pairs of variables is conditionally independent. For example,

   x is independent of *z*, controlling for *y*;

   x and *y* are associated, controlling for *z*;

   y and *z* are associated, controlling for *x*.

A Hierarchy of Loglinear Models for Three Variables

4. No pair of variables is conditionally independent, but the association between any two variables is the same at each category of the third. We then say there is ***homogeneous association***.
5. All pairs of variables are associated, but there is *interaction*; that is, the association between each pair varies according to the category of the third variable.

A Hierarchy of Loglinear Models for Three Variables

- Each model has a symbol that indicates the pairs of variables that are associated.
- (xy, z) denotes the model for case 2 (above) in which x and y are associated but the other two pairs are conditionally independent.
- The symbol (xy, xz, yz) denotes the model for case 4, in which all three pairs are associated but the association is homogeneous.

A Hierarchy of Loglinear Models for Three Variables

- All the models provide some structure for the pattern of association except for the one symbolized by (*xyz*). This model fits any sample three-way table perfectly, allowing the associations to be nonhomogeneous. It is called the ***saturated model***.

TABLE 15.13: Some Loglinear Models for Three-Dimensional Contingency Tables

Model Symbol	Interpretation
(x, y, z)	All pairs are conditionally independent.
(xy, z)	x and y are the only associated pair.
(xy, yz)	x and z are the only conditionally independent pair.
(xy, yz, xz)	Each pair is associated, controlling for the third variable, but the association is homogeneous.
(xyz)	All pairs are associated, but the association is nonhomogeneous (interaction).

Odds Ratio Interpretations for Loglinear Models

- Interpretations of associations in loglinear models, like those in logistic regression models, can use the odds ratio.
- In a three-way table, ***conditional independence*** between x and y means that the population odds ratios in the xy partial tables all equal 1.0.
- ***Homogeneous association*** means that the population odds ratios in the xy partial tables are identical at each category of z.

Example: Estimated Odds Ratios for Substance Use Data

Denote the variables in the first example by *A* for alcohol use, *C* for cigarette use, and *M* for marijuana use. The next table contains estimated expected frequencies for the loglinear model (*AC,AM,CM*) that permits an association for each pair of variables but assumes homogeneous association, with the odds ratio between two variables being the same at each level of the third variable. The estimated expected frequencies are very close to the observed counts, so the model seems to fit well.

Example: Estimated Odds Ratios for Substance Use Data

TABLE 15.14: Estimated Expected Frequencies for the Loglinear Model (*AC*, *AM*, *CM*) for Alcohol (*A*), Cigarette (*C*), and Marijuana (*M*) Use for High School Seniors

Alcohol Use	Cigarette Use	Marijuana Use	
		Yes	No
Yes	Yes	910.4	538.6
	No	44.6	455.4
No	Yes	3.6	42.4
	No	1.4	279.6

Example: Estimated Odds Ratios for Substance Use Data

Let's study the estimated association between cigarette use and marijuana use, controlling for alcohol use, using the estimated expected frequencies. For those who have used alcohol, the estimated odds ratio between *C* and *M* is

$$\frac{910.4 \times 455.4}{538.6 \times 44.6} = 17.3.$$

Similarly, for those who have not used alcohol, the estimated odds ratio between *C* and *M* is

$$\frac{3.6 \times 279.6}{42.4 \times 1.4} = 17.3.$$

Example: Estimated Odds Ratios for Substance Use Data

For each category of *A*, students who have smoked cigarettes have estimated odds of having smoked marijuana that are 17.3 times the estimated odds for students who have not smoked cigarettes. The model assumes homogeneous association, so the estimated odds ratio is the same at each category of *A*.

Example: Estimated Odds Ratios for Substance Use Data

Software for loglinear models provides tables of model parameter estimates from which one can also find estimated odds ratios. The next table illustrates this. For each pair of variables, the association parameter estimate refers to the *log* odds ratio. For the *CM* conditional association, therefore, the estimated odds ratio at each level of *A* equals $e^{2.848} = 17.3$. Similarly, the estimated odds ratio equals $e^{2.054} = 7.8$ between *A* and *C* at each level of *M*, and the estimated odds ratio equals $e^{2.986} = 19.8$ between *A* and *M* at each level of *C*. The estimated conditional association is very strong between each pair of variables.

Example: Estimated Odds Ratios for Substance Use Data

TABLE 15.15: Output of Association Parameter (Log Odds Ratio) Estimates for the Loglinear Model (*AC*, *AM*, *CM*) for Substance Use Data

Parameter	Estimate	Std. Error	z	Sig.
A*C	2.0545	0.1741	11.80	.000
A*M	2.9860	0.4647	6.43	.000
C*M	2.8479	0.1638	17.38	.000

Odds Ratio Interpretations for Loglinear Models

- The model (*AC,AM,CM*) permits conditional association for each pair of variables.
- Other possible loglinear models for these data delete at least one of the associations.
- To illustrate the association patterns implied by some of these models, The next table presents estimated conditional odds ratios for the estimated expected frequencies for the models.

Odds Ratio Interpretations for Loglinear Models

TABLE 15.16: Summary of Estimated Conditional Odds Ratios for Various Loglinear Models Fitted to Substance Use Data

	Conditional Odds Ratio		
Model	AC	AM	CM
(A, C, M)	1.0	1.0	1.0
(AC, M)	17.7	1.0	1.0
(AM, CM)	1.0	61.9	25.1
(AC, AM, CM)	7.8	19.8	17.3
(ACM) Level 1	13.8	24.3	17.5
(ACM) Level 2	7.7	13.5	9.7

15.7 Model Goodness-of-Fit Tests for Contingency Tables

- A ***goodness-of-fit test*** for a model is a test of the null hypothesis that that model truly holds in the population of interest.

Chi-Squared Goodness-of-Fit Statistics

- Each model for a contingency table has a set of cell estimated expected frequencies.
- The model goodness of fit is tested by comparing the estimated expected frequencies, denoted by $\{f_e\}$, to the observed frequencies $\{f_o\}$.
- Chi-squared test statistics summarize the discrepancies.
- Larger differences between the $\{f_o\}$ and $\{f_e\}$ yield larger values of the test statistics and stronger evidence that the model is inadequate.

Chi-Squared Goodness-of-Fit Statistics

- Two chi-squared statistics, having similar properties, can test goodness of fit. The ***Pearson chi-squared statistic*** is

$$X^2 = \sum \frac{(f_o - f_e)^2}{f_e}.$$

- Another statistic, the ***likelihood-ratio chi-squared statistic***, is

$$G^2 = 2 \sum f_o \log \left(\frac{f_o}{f_e} \right).$$

Chi-Squared Goodness-of-Fit Statistics

- If the model truly holds, both test statistics have approximate chi-squared distributions.
- The chi-squared approximation is better for larger sample sizes.

Example: Logistic Model Goodness of Fit for Death Penalty Data

For the death penalty verdicts (the table is shown again on the next slide), the examples used the model

$$\text{logit}[P(y = 1)] = \alpha + \beta_1 d + \beta_2 v$$

to describe how the probability of receiving the death penalty depends on defendant's race *d* and victims' race *v*. A goodness-of-fit test analyzes whether this model with main effects is adequate for describing the data. The more complex model containing the interaction term is necessary if the main effects model fits poorly.

Example: Logistic Model Goodness of Fit for Death Penalty Data

TABLE 15.3: Death Penalty Verdict by Defendant's Race and Victims' Race, for Cases with Multiple Murders in Florida

Defendant's Race	Victims' Race	Death Penalty Yes	Death Penalty No	Percentage Yes
White	White	53	414	11.3
	Black	0	16	0.0
Black	White	11	37	22.9
	Black	4	139	2.8

Example: Logistic Model Goodness of Fit for Death Penalty Data

Software automatically finds the model's estimated expected frequencies and the goodness-of-fit statistics. For instance, this model estimated a probability of 0.233 that a black defendant receives the death penalty for having white victims. The data showed there were 48 such black defendants, so the estimated expected number receiving the death penalty is 48(0.233) = 11.2. This is the estimated expected frequency for the cell in the table having observed frequency 11.

Example: Logistic Model Goodness of Fit for Death Penalty Data

The *df* equal the number of logits minus the number of parameters in the model. The death penalty data have four logits, one for each combination of defendant's race and victims' race. The model has three parameters, so both goodness-of-fit statistics have

df = Number of logits − Number of parameters = 4 − 3 = 1.

Example: Logistic Model Goodness of Fit for Death Penalty Data

The table on the next slide shows software results of chi-squared goodness-of-fit tests for the logistic model for the death penalty data. The null hypothesis for the tests is that the logistic model with main effects truly holds; that is, no interaction occurs between defendant's race and victims' race in their effects on the death penalty verdict. The Pearson test statistic is $X^2 = 0.20$ and the likelihood-ratio test statistic is $G^2 = 0.38$. These test statistic values are small, so neither P-value is small. The model fits the data well. The null hypothesis that the model holds is plausible.

Example: Logistic Model Goodness of Fit for Death Penalty Data

TABLE 15.17: Chi-Squared Goodness-of-Fit Tests for Logistic Model with Main Effects Fitted to Death Penalty Data (Table 15.3)

	Goodness-of-fit Tests		
	Value	df	Sig.
Likelihood Ratio	.380	1	.538
Pearson Chi-Square	.198	1	.656

Standardized Residuals

- The chi-squared goodness-of-fit statistics provide global measures of lack of fit.
- When the fit is poor, a closer look at the cells of the table may reveal the nature of the lack of fit.
- Software for logistic and loglinear models can report ***standardized residuals***, which make a cell-by-cell comparison of f_o and f_e. Each standardized residual has the form

$$\text{Standardized residual} = \frac{f_o - f_e}{\text{Standard error of } (f_o - f_e)}.$$

Standardized Residuals

- When the model truly holds, standardized residuals behave like standard normal variables.
- For the logistic model for the death penalty data, the standardized residuals all equal ± 0.44.
- They are small and provide no evidence of lack of fit. This is not surprising, since the goodness-of-fit statistics are small.

Loglinear Model Goodness of Fit

- The same goodness-of-fit formulas apply to loglinear models.
- Likewise, standardized residuals compare individual cell counts to expected frequencies satisfying the model.

Loglinear Model Goodness of Fit

- From the table on the next slide, the only model that passes the goodness-of-fit test is (*AC*,*AM*,*CM*).
- This model allows association between all pairs of variables but assumes that the odds ratio between each pair is the same at each category of the third variable.
- The models that lack any associations fit poorly, having *P*-values of 0.000.

Loglinear Model Goodness of Fit

TABLE 15.18: Goodness-of-Fit Tests for Loglinear Models of Alcohol (A), Cigarette (C), and Marijuana (M) Use, with Likelihood-Ratio (G^2) and Pearson (X^2) Chi-Squared Test Statistics

Model	G^2	X^2	df	P-Value
(A, C, M)	1286.0	1411.4	4	0.000
(A, CM)	534.2	505.6	3	0.000
(C, AM)	939.6	824.2	3	0.000
(M, AC)	843.8	704.9	3	0.000
(AC, AM)	497.4	443.8	2	0.000
(AC, CM)	92.0	80.8	2	0.000
(AM, CM)	187.8	177.6	2	0.000
(AC, AM, CM)	0.4	0.4	1	0.54

Comparing Models by Comparing G^2-Values

- The prior table illustrates two important properties of the likelihood-ratio G^2 statistic.
- First, G^2 has similar properties as the SSE measure in regression. Both compare observed responses to values expected if a model holds, and both cannot increase as the model becomes more complex.
- Second, to test the null hypothesis that a model truly holds versus the alternative hypothesis that a more complex model fits better, the test statistic is the difference in G^2-values.

Distinction Between Logistic and Loglinear Models

- Logistic regression models distinguish between a single response variable and a set of explanatory variables.
- By contrast, loglinear models treat *every* variable as a response variable.

15.8 Chapter Summary

- The methods of this chapter showed how to model a categorical response variable in terms of possibly *several* explanatory variables, which can be categorical or quantitative or both.

Chapter Summary

- For binary response variables, the ***logistic regression*** model describes how the probability of a particular category depends on explanatory variables. It uses a linear model for the ***logit*** transform of the probability, which is the log of the odds. For a quantitative explanatory variable, an S-shaped curve describes how the probability changes as the explanatory variable changes.

Chapter Summary

- The antilog of a $\hat{\beta}$ parameter estimate in logistic regression is a multiplicative effect on the odds for the response variable, for each one-unit increase in the explanatory variable of which it is a coefficient. Thus, for logistic regression the ***odds ratio*** is a natural measure of the nature and strength of an association.

Chapter Summary

- A parameter value of $\beta = 0$ corresponds to a predictor having no effect on the response. To test H_0: $\beta = 0$, we can use the normal test statistic, $z = \hat{\beta}/se$. The ***Wald test*** uses the square of this ratio. The ***likelihood-ratio test*** compares values of $(-2 \log \ell)$ for models with and without that term, where is the maximized *likelihood* function. The large-sample distribution of these test statistics is chi-squared with $df = 1$. The likelihood-ratio statistic can also test H_0: $\beta_1 = \cdots = \beta_p = 0$ in multiple logistic regression (with $df = p$) or compare nested models.

Chapter Summary

- For *ordinal* response variables, an extension of logistic regression uses *cumulative logits*, which are logits of *cumulative probabilities*. The model is called a ***cumulative logit model***. The effects of the explanatory variables are the same for each cumulative probability.
- For *nominal* response variables, an extension of logistic regression forms logits by pairing each category with a baseline category. Each logit equation has separate parameters in this ***baseline-category logit model***.

Chapter Summary

- ***Loglinear*** models are useful for investigating association patterns among a set of categorical response variables. They consider possible conditional independence patterns and use conditional odds ratios to describe association.
- For models for contingency tables, Pearson and likelihood-ratio chi-squared statistics test the ***goodness of fit*** of models to the data.